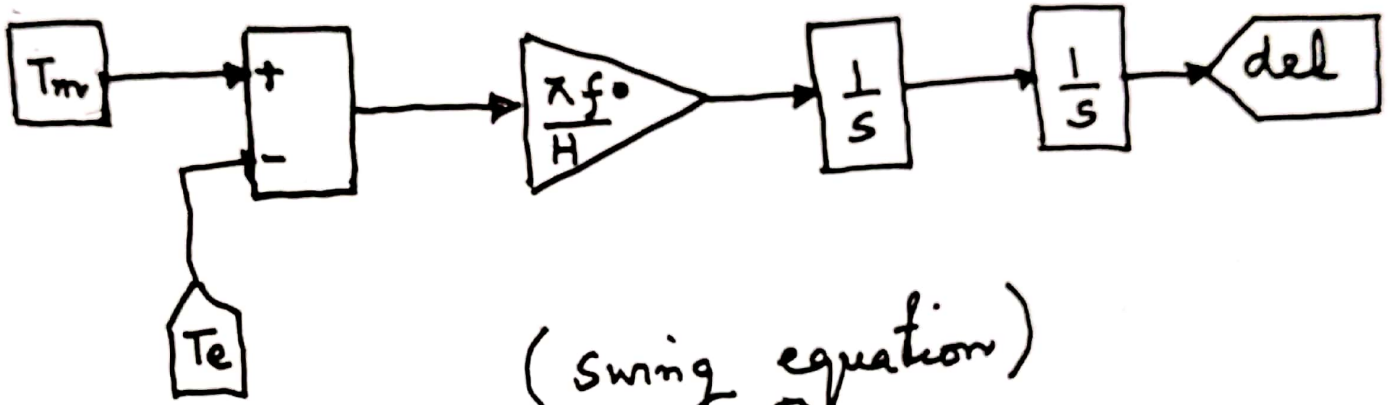
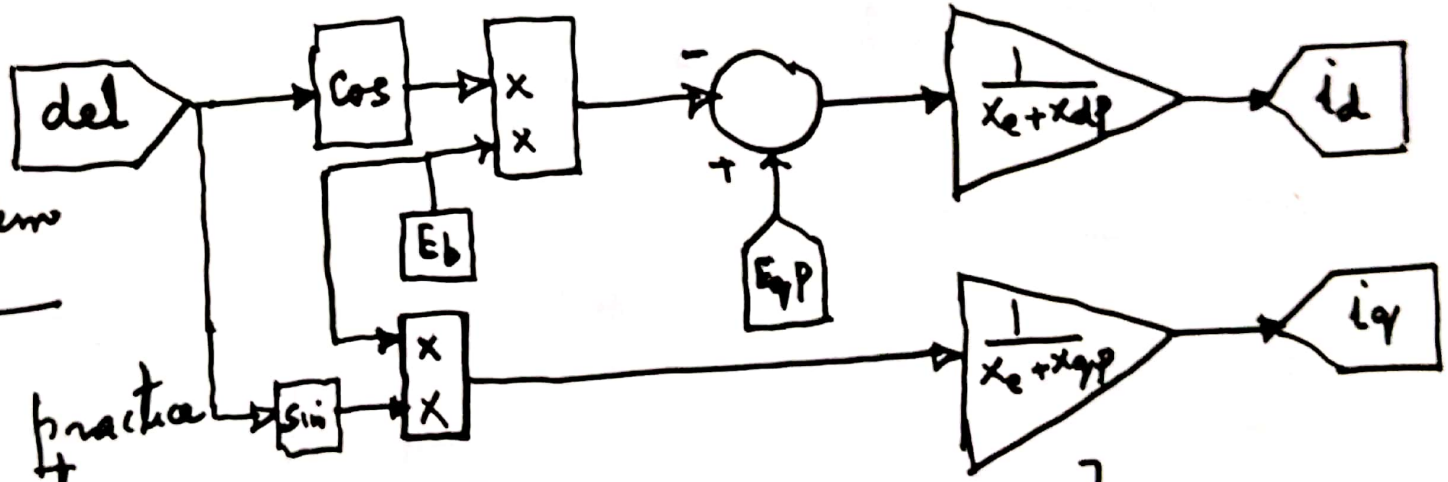
Base Case (SMIB) System



(swing equation ①)

GoTo
From

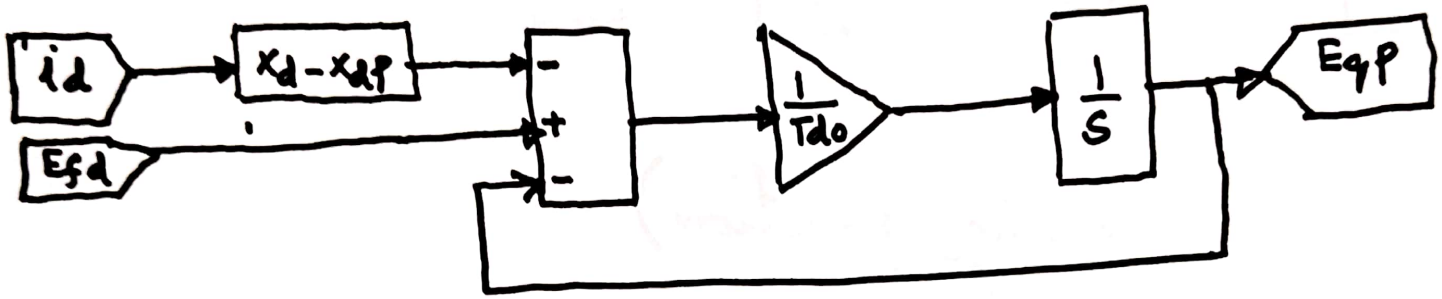
All integrator realization of SMIB system



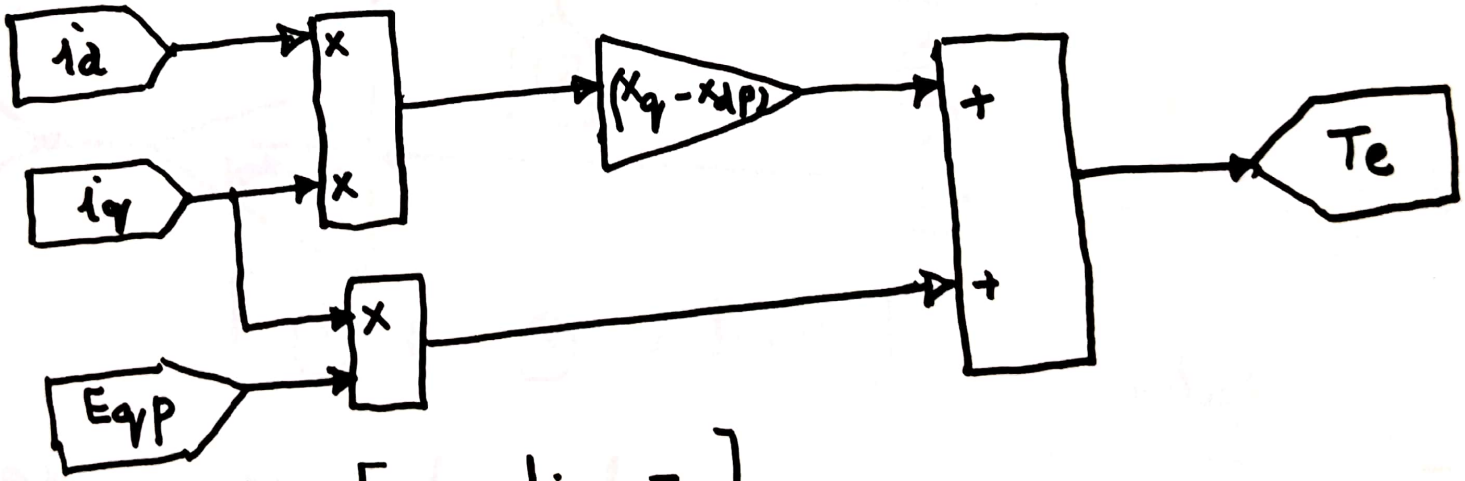
[equations ② & ③]

Standard practice in power system dynamic formulation

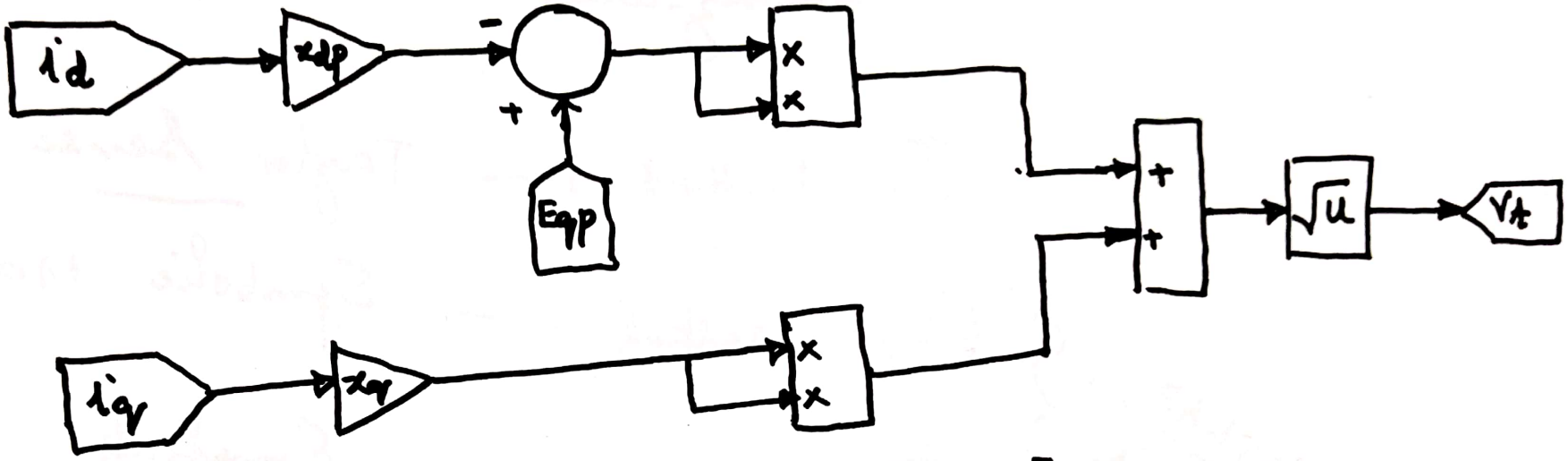
Initial conditions of integrators are assigned directly from the pre-disturbance operating point



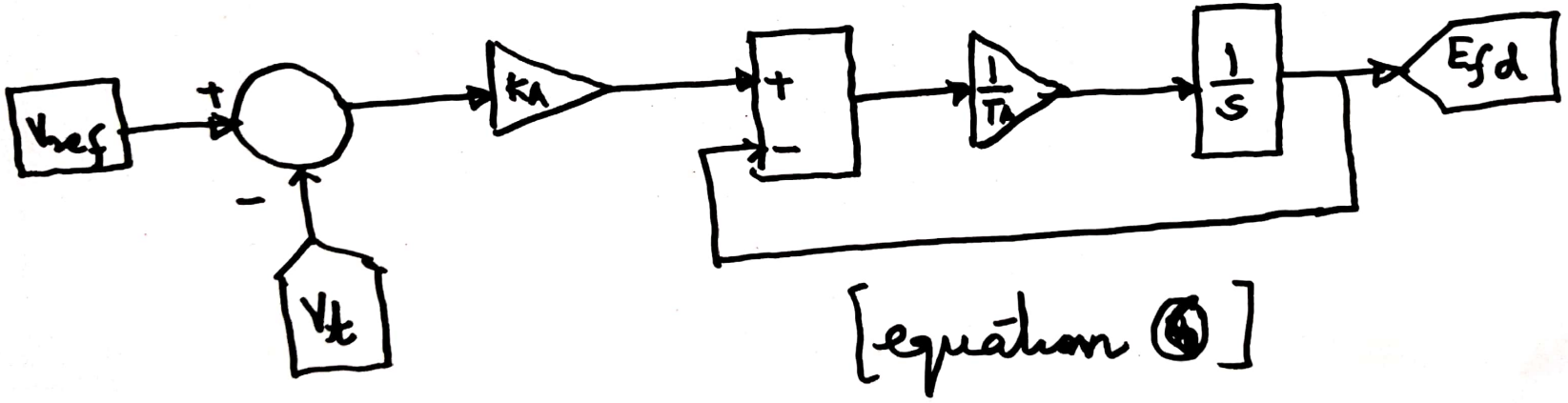
[equation 4]



[equation 5]



[equations ④, ⑤ & ⑥]



[equation ⑦]

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Linearization

1. Method 1 — Taylor Series (Manual)
2. Method 2 — Symbolic Math Toolbox
3. Method 3 — Simulink Linearization

MATLAB / Simulink

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